

18 Ans: B

A : can't be correct because when the velocity = 0 at $t = 0$, the KE (Q) should also be zero (instead of at a maximum)

C : can't be correct because $PE = \frac{1}{2} m \omega^2 x^2$. Thus at $t = 0$, when displacement (P) = 0, PE (Q) should also be 0 (instead of at a max)

D : is incorrect because at $t = 0$, when net force (P) = 0, displacement $x = 0$ since $F=ma = m \omega^2 x$

Thus when $x = 0$, its PE (Q) cannot be a maximum since $PE = \frac{1}{2} m \omega^2 x^2$.

B : When acceleration = 0, at $t = 0$, $x = 0$ also since $a = -\omega^2 x$

When $x = 0$, velocity = max, since $v = \omega \sqrt{x_o^2 - x^2}$; thus KE (Q) = max.